

REAL ESTATE BY THE NUMBERS

The Ultimate Numbers Reference Guide

#1: **Net worth** = Total value of assets – Total value of liabilities (debts)

#2: **Assets** are resources owned or controlled by you or your business that can be measured in dollars and will provide some future value.

#3: **Liquid assets** are assets that are easily turned into cash, such as stocks, bonds, and—of course—actual cash.

#4: **Appreciating assets** are assets that tend to increase in value over time. **Depreciating assets** are assets that tend to decrease in value over time.

#5: **Liabilities** are legal debts and obligations that reduce future financial value.

#6: **The Personal Financial Statement (PFS) Equation:**

$$\text{Net Worth} = \text{Assets} - \text{Liabilities}$$

#7: **Savings rate** = Monthly income – Monthly expenses

#8: **Investable Assets** = Liquid Assets – (3 × Monthly Expenses)

#9: **The balance sheet** is a snapshot in time of everything your business owns, everything it owes, and the difference between those two numbers (which is the total value of the business).

#10: **Owner equity** is the total value of the company to the owners.

#11: **The Balance Sheet Equation:**

$$\text{Assets} = \text{Liabilities} + \text{Owner equity}$$

#12: **The income statement** (also known as the **profit and loss statement**) is a breakdown of revenue, expenses, and total profitability for a business or investment over a defined period of time.

#13: **Income** is a general term referring to the money flowing into a business or investment. Sometimes you will see income referred to as “revenue” or “sales” instead.

#14: **Cost of goods sold (COGS)**, also called **cost of sales**, is the direct cost to produce the goods and services sold by a company.

#15: **Gross profit** is the amount of income a company makes after subtracting the costs associated with making and selling its products and services from its total revenue.

$$\text{Gross profit} = \text{Income} - \text{Cost of goods sold}$$

#16: **Operating expenses** are the costs incurred in the normal operations of the business.

#17: **Operating income** is the amount of profit a company makes after subtracting both gross profit and operating expenses from revenue.

$$\text{Operating income} = \text{Gross profit} - \text{Operating expenses}$$

#18: **Net income** is the amount of profit a company makes after subtracting cost of goods sold, operating expenses, taxes, and interest payments from the total revenue.

$$\text{Net income} = \text{Operating income} - \text{Taxes} - \text{Interest}$$

#19: **Profit margin** is the ratio of the profit a business generates to the amount of revenue brought in. The more money a business keeps in profit, the higher its profit margins.

$$\text{Profit margin} = \text{Profits} \div \text{Revenue}$$

#20: **Gross profit margin** is the ratio of the amount of gross profit a business generates to the amount of revenue brought in. This ratio indicates how efficiently the company is producing their products compared to their competitors.

$$\text{Gross profit margin} = \text{Gross profit} \div \text{Revenue}$$

#21: **Operating profit margin** is the ratio of the amount of operating income a business generates to the amount of revenue brought in. This ratio indicates how efficiently the company is operating, including both product creation and management.

$$\text{Operating profit margin} = \text{Operating income} \div \text{Revenue}$$

#22: **Net profit margin** is the ratio of the amount of net income a business generates to the amount of revenue brought in. This ratio indicates how efficiently the company is operating overall.

$$\text{Net profit margin} = \text{Net income} \div \text{Revenue}$$

#23: **Gross operating income (GOI)**

$$\text{GOI} = \text{Gross potential rent} - \text{Rent loss} + \text{Other income}$$

#24: **Gross potential rent**

$$\text{Gross potential rent} = \text{Number of units} \times \text{Market rent} \times 12$$

#25: **Operating expenses** (in real estate) are expenses associated with the basic functions and operations of the property, such as insurance, repairs and maintenance, and taxes.

#26: **Net operating income (NOI)** is the gross operating income of a property minus operating expenses.

$$\text{NOI} = \text{Gross operating income} - \text{Operating expenses}$$

#27: **Cash flow** is the amount of pre-tax cash a property generates. It is equal to the net operating income minus debt service and capital expenses.

$$\text{Cash flow} = \text{NOI} - \text{Debt service} - \text{Capital expenses}$$

#28: **Interest** is the periodic payment against money that is borrowed or lent.

#29: **Simple interest**

$$\text{Interest} = \text{Principal} \times \text{Interest rate}$$

#30: **Total interest**

$$\text{Total interest} = \text{Principal} \times \text{Interest rate} \times \text{Number of years}$$

#31: **Interest rate**

$$\text{Interest rate} = \text{Interest} \div \text{Principal}$$

#32: **Compound interest**

$$\text{Compound interest} = (\text{Principal} \times (1 + \text{Interest rate})^{\text{Periods}}) - \text{Principal}$$

#33: Expected value

$$EV = (E^1 \times P^1) + (E^2 \times P^2) + \dots + (E^N \times P^N)$$

Where

E = The expected financial result of the outcome, and

P = The probability of the outcome.

#34: Time value of money (TVM) is the concept that money in hand today is worth more than money in hand at some future time.

#35: Future value formula

$$FV = PV \times (1 + i)^n$$

#36: Discount rate is the rate of return used to discount future value (or future cash flows) back to their present value.

#37: Present value formula

$$PV = FV \div (1 + i)^n$$

#38: Discounted cash flow (DCF) gives you the present value of an entire stream of income.

$$DCF = (CF_1 \div (1 + i)^1) + (CF_2 \div (1 + i)^2) + \dots + (CF_x \div (1 + i)^x)$$

where

CF is each of the future cash flows, and

i is the discount rate.

#39: Net present value (NPV) compares the current value of all future cash flows generated by a project while considering the initial capital investment, in order to determine whether the investment will be profitable.

#40: Internal rate of return (IRR) is the discount rate that makes NPV = 0 for an investment; IRR is the compounded return of that investment.

#41: Taxable income

$$\text{Taxable income} = \text{NOI} - \text{Mortgage interest} - \text{Depreciation} - \text{Amortization}$$

#42: Depreciation is a tax adjustment that accounts for the declining condition of a physical property. It allows real estate investors to delay a portion of their income taxes for a given property until that property is sold.

#43: **Amortization** refers to tax deductions against loan expenses that are taken over a period of time, as opposed to all at once.

#44: **Basis** (or “**cost basis**”) is the starting point for the tax liability associated with selling a property. It is generally defined as the purchase price plus any closing costs.

#45: **Adjusted basis** is the end point for the tax liability associated with selling a property. It adjusts the “basis” by accounting for capital expenditures, sales costs, and depreciation.

#46: **Adjusted basis formula**

Adjusted basis = Purchase price + Purchase costs + Capital expenditures + Selling costs – Depreciation

#47: **Taxable gain/Loss at sale**

Taxable gain/Loss at sale = Sale price – Adjusted basis

#48: **Cash flow before taxes**

Cash flow before taxes = NOI – Debt service – Capital expenses + Loan additions

#49: **Cash flow after taxes**

Cash flow after taxes = Cash flow before taxes – Income tax

#50: **Return on investment (ROI)**

ROI = (Ending value – Starting value) ÷ (Starting value)

#51: **Annualized ROI**

Annualized ROI = ROI ÷ Years held

#52: **Equity multiplier (EM)**

EM = Ending value ÷ Starting value

#53: **Capitalization rate (cap rate)**

Cap rate = NOI ÷ Value

#54: **Cash-on-cash return (COC)**

COC = Annual cash flow ÷ Cash invested

#55: **Average annual return (AAR)**

AAR = (ROI₁ + ROI₂ + ... + ROI_N) ÷ Years held

#56: Compound annual growth rate (CAGR)

$$\text{CAGR} = (\text{Ending value} \div \text{Starting value})^{(1 \div n)} - 1$$

Where n = number of periods

#57: Gross rent multiplier (GRM)

$$\text{GRM} = \text{Value} \div \text{Gross potential income}$$

#58: Financing refers to all the capital used to purchase an asset, like a business or a real estate investment. Financing can fall into different categories (e.g., debt, equity) and can be provided or obtained from a variety of sources.

#59: Preferred equity holders are given preference in the capital stack over common equity holders. Typically, they will receive returns up to a specific amount or percentage (the “preferred return”) before common equity holders receive any return. Preferred equity holders have more risk than debt holders but less risk than common equity holders.

#60: Debt financing involves borrowing money in the form of a loan and then repaying that money with a predefined amount of interest.

#61: Principal

$$\text{Principal} = \text{Purchase price} \times (1 - \text{Down payment \%})$$

#62: Monthly mortgage payments

$$M = P \times [(r (1 + r)^n \div ((1 + r)^n) - 1]$$

Where

M = monthly mortgage payment

P = total loan amount

r = monthly interest rate (Divide your annual interest rate by 12. For a 5 percent interest rate, r = 0.417 percent)

n = total number of payments (Multiply the term of your loan in years by 12. For a thirty-year mortgage, n = 360)

#63: Equity value

$$\text{Equity value} = \text{Property value} - \text{Liabilities}$$

#64: Loan-to-value ratio (LTV)

$$\text{LTV} = \text{Loan amount} \div \text{Property value}$$

#65: Loan-to-cost ratio (LTC)

$$\text{LTC} = \text{Loan amount} \div \text{All-in costs}$$

#66: Debt service coverage ratio (DSCR)

$$\text{DSCR} = \text{Net operating income} \div \text{Total debt service}$$

#67: The loan constant (or mortgage constant) is a ratio that indicates the relationship between a loan and its monthly payment. The ratio (percentage) is the cash paid to service a loan in one year divided by the total amount owed on the loan.

$$\text{Loan constant} = \text{Annual debt service} \div \text{Loan amount}$$

#68: Breakeven loan value is the amount you can borrow using a specific loan for a specific property where the cash flow transitions from positive to negative.

$$\text{Breakeven loan value} = \text{NOI} \div \text{Loan constant}$$

#69: Leverage (the concept) is the use of borrowed capital for an investment with the intent to generate a return on the borrowed capital that will be greater than the interest paid to secure the capital.

#70: Positive leverage occurs when the use of borrowed capital generates higher returns on an investment than would be attained by paying for that investment with non-borrowed funds.

#71: Negative leverage occurs when the use of borrowed capital reduces the returns on an investment compared to if that investment were purchased with non-borrowed funds.

#72: Leverage (the value) is the difference between the cap rate of an investment and the loan constant of a particular financing option for that investment.

$$\text{Leverage} = \text{Cap rate} - \text{Loan constant}$$

#73: Determination of positive or negative leverage is done by looking at the Leverage value. If Leverage is a positive number (the cap rate is higher than the loan constant), the loan will provide positive leverage. If Leverage is a negative number (the cap rate is lower than the loan constant), the loan will provide negative leverage.

#74: Return on equity (ROE)

$$\text{ROE} = \text{Cash flow} \div \text{Equity value}$$

#75: Profit

$\text{Profit} = \text{Sales price} - \text{Purchase price} - \text{Expenses}$

Where

Sales price is the conservative estimate of what you can sell the property for at sale.

Purchase price is the expected cost to acquire the property.

Expenses are the expected total costs associated with buying, renovating, holding, and then reselling the property.

#76: Equity-on-cash return (EOC)

$\text{EOC} = \text{Annual principal paydown} \div \text{Cash invested}$

#77: Annual depreciation

$\text{Annual depreciation} = \text{Building value} \div 27.5$

#78: Tax savings

$\text{Tax savings} = \text{Taxable income deduction} \times \text{Marginal tax rate}$

THE METRIC MATRIX

- Q1. How much money will I make from this investment?
- Q2. What percentage of my initial investment will this investment return?
- Q3. What percentage of my initial investment will this investment return per year?
- Q4. What is the compounded return of this investment over time?

RETURN METRIC	ANSWERS Q1	ANSWERS Q2	ANSWERS Q3	ANSWERS Q4
Profit	X			
Net Operating Income (NOI)	X			
Cash Flow (CF)	X			
Return on Investment (ROI)		X		
Equity Multiplier (EM)		X		
Capitalization Rate (Cap Rate)			X	
Cash-on-Cash Return (COC)			X	
Average Annual Return (AAR)			X	
Compound Annual Growth Rate (CAGR)				X
Internal Rate of Return (IRR)				X